

Transport Laws in Closure Dynamics

Discrete continuity, $U(1)$ current conservation, and the photon-sector zero mode on V_{600}

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Abstract

We collect the transport-law content of the VFD closure programme into a single conditional framework, organised around the closure functional $\mathcal{F}[\psi] = \alpha R[\psi] + \beta E[\psi] - \gamma C[\psi]$ (with $C[\psi]$ the quadratic closure term and the symbol Q reserved elsewhere for the total-probability charge) and the 600-cell spectral substrate of P2 §6 [12]. The paper is deliberately packaged as a *unifying proposal with explicit hypothesis audit*, not as a closed theorem chain.

The paper contains: (i) a symmetry catalogue for $\mathcal{F}$ with *two* exact symmetries ($U(1)$ phase, H_4 isometry) and two approximate ones (σ -Galois, $SO(4)$ -rotation) explicitly flagged; (ii) a discrete 600-cell continuity law stated as a *programme target* (Hypothesis 3.1), whose generator-level derivation from the closure Langevin dynamics is flagged as an open item — Paper XVIII’s continuum continuity is imported, and the discrete analogue is targeted, not proved, here; (iii) one conditional conservation statement — the $U(1)$ total-probability conservation under that continuity hypothesis (Proposition 4.1) — and one H_4 -equivariance remark recording a conditional block-norm-invariance statement — valid only under the Schrödinger-limit reduction of the Madelung pair *plus* the additional hypothesis that the Schrödinger-limit generator $\mathcal{H}$ is itself H_4 -equivariant (no continuous angular-momentum current is claimed; H_4 is a finite Coxeter group without a Lie algebra); (iv) a Schrödinger-limit dispersion identification (Hypothesis 5.1) $\omega^2 = c_{\text{eff}}^2 k^2 + m_\rho^2$ with $m_\rho^2 = \beta \lambda_\rho$, λ_ρ the Laplacian eigenvalue of P2 §6, presented as a heuristic physical identification rather than a theorem; (v) a proposed dictionary between the transport-law objects and the ARIA cognitive substrate (**aria-chess** repository), stated as a proposal pending repository-backed verification; and (vi) a hypothesised specialisation to photon transport ($\lambda = 0$ massless eigenmode) conditional on the T_{PH} chain recorded in the internal note [14], with the $T_{\text{PH}2}$ “needs polish” caveat of that note reproduced explicitly in the claim.

Microtubule transport is reframed: the paper explicitly does *not* derive the 13-fold helical symmetry from L_{12} . The claim attached to microtubules is a hypothesis — that biological microtubules *may be* active-transport substrates of ARIA’s class, with 13 as a selected operating point — not a theorem.

Epistemic status. Proposal + hypothesis audit, with two new theorems delivered in the Schrödinger-limit regime of the closure dynamics on V_{600} :

- Theorem 4.3 (*discrete continuity + $U(1)$ conservation, with explicit current*): for the Schrödinger-limit evolution $i\dot{\psi} = H\psi$ with H any real-symmetric *nearest-neighbour-supported* operator on $\mathbb{R}^{V_{600}}$ (the graph Laplacian $L = D - A$ is the canonical example), the explicit edge current $j_{v \rightarrow w} := -2 \text{Im}(\overline{\psi_v} H_{vw} \psi_w)$ satisfies the discrete continuity equation pointwise, is antisymmetric, and yields exact $U(1)$ total-probability conservation. This promotes Hypothesis 3.1 and Proposition 4.1 from conditional to unconditional in the Schrödinger limit on $H = L$.
- Theorem 5.3 (*photon-sector spectral witness*): the $\lambda = 0$ eigenspace of $L_{V_{600}}$ is one-dimensional, lies in the trivial $2\mathbf{I}$ -isotypic sector, and verifies the substrate-side zero-

mode facts used by the massless ($m = 0$) specialisation of the dispersion ansatz of Hypothesis 5.1; non-trivial eigenmodes oscillate at $\omega = \lambda$ exactly.

The five remaining open items (gauge-field emergence; Γ -limit coarse-graining; learned- W -as-gauge-potential; ARIA repository-level verification (*reduced scope: the active-regime witness is now the published [aria-chess](#) preprint [2]*); a generator-level derivation of the discrete continuity law from the closure Langevin process (*Schrödinger-limit case now closed; Langevin-with-noise case still open*)) are listed in §10. Both new theorems are independently reproducible from canonical numerical scripts in [papers/transport-law/repro/](#) (§11; wall-clock seconds on a laptop, no fitted parameters; the analytic continuity residual is at machine precision, $\sim 3 \times 10^{-16}$).

Status of claims at a glance.

Claim type	Content	Status
Theorem	Schrödinger-limit explicit current $j_{v \rightarrow w} = -2 \text{Im}(\psi_v H_{vw} \psi_w)$, antisymmetry, pointwise discrete continuity, exact $U(1)$ total-probability conservation (Theorem 4.3)	<i>proved here</i>
Theorem	V_{600} Laplacian zero-mode is one-dimensional, lies in trivial $2\mathbf{I}$ -isotypic sector, is stationary under $i\psi = L\psi$; non-trivial modes rotate at $\omega = \lambda$ (Theorem 5.3)	<i>proved here</i>
Hypothesis	Discrete continuity law for the full <i>Langevin-with-noise</i> closure dynamics (Hypothesis 3.1)	<i>open</i>
Hypothesis	Relativistic-form dispersion identification $\omega^2 = c_{\text{eff}}^2 k^2 + m_\rho^2$ with $m_\rho^2 = \beta \lambda_\rho$ (Hypothesis 5.1)	<i>heuristic identification</i>
Hypothesis	Identification of $\lambda = 0$ eigenmode with the physical $U(1)_{\text{EM}}$ photon sector (Hypothesis 7.1)	<i>conditional on T_{PH}</i>
Programme target	ARIA <code>couple_tick</code> verification of Fick / Nernst-Planck / cascade-pressure-balance objectives (§6)	<i>not done here</i>
Programme target	Microtubule active-substrate biophysics (Hypothesis 8.1)	<i>future biophysics</i>

Contents

1	Introduction	3
1.1	The programme gap this paper closes	3
1.2	Structure	3
1.3	Notation and conventions	4
2	The closure functional and its symmetries	4
3	The master continuity equation	5
3.1	Discrete continuity on V_{600}	5
3.2	Continuum limit (conjectural)	6
4	Symmetry-associated conservation statements (Noether caveats)	6
4.1	Noether caveat	6
5	Dispersion on the 600-cell	9
6	ARIA as a proposed verification-target testbed (not done here)	10
6.1	Mapping	10
6.2	Objectives 2, 4, 6 as proposed ARIA verification targets	11
6.3	Numerical verification plan	12

7	Massless zero mode as a conditional photon-sector witness	12
8	Microtubule transport: reframed	12
8.1	Reframed claim	12
9	Consistency checks with Papers XII–XXXIV	13
10	Open items	14
11	Numerical verification	14
12	Conclusion	15

1 Introduction

1.1 The programme gap this paper closes

The VFD crystallisation programme derives static spectral observables — particle masses (Paper V), charge radii (Paper XXXII), the fine-structure constant as a structural correspondence (Paper XXXIV) — and event-order kinematics on the GR side (Papers XXIII–XXVII). External reviewers have flagged that no single paper catalogues the *dynamical* content of the framework. The present paper’s *delivered* content is narrower than that full programme: we close the probability-current layer in the Schrödinger limit (Theorem 4.3, with the discrete continuity equation, antisymmetric current, and $U(1)$ conservation all unconditional for $H = L$), leave the Langevin-with-noise derivation conditional, and record analogies / hypotheses / proposals for the remaining transport content (energy flow, linear-momentum flow, cascade-pressure flow). Dynamical charge flow, energy flow, linear-momentum flow, and cascade-pressure flow are discussed only under explicit hypothesis load, and the paper is explicit about where each hypothesis fails to close. Transport structure exists in fragmentary form across

- Paper XII (gradient flow $\dot{\psi} = -\nabla\mathcal{F}$),
- Paper XVI (WKB transport; characteristic flow along $\nabla\mathcal{F}$),
- Paper XVIII (Nelson pairing; exact continuity equation; Madelung pair),
- Paper XIX (nonlinear closure residual),
- Paper XXVII (discrete conservation $\sum_v(S_v - \bar{S}) = 0$),
- Paper XXX (stationary measure).

This paper collects those ingredients into a hypothesis-audited framework. It closes the conditional probability-current layer; beyond that, it catalogues the two exact symmetries of $\mathcal{F}[\psi]$, records a heuristic dispersion identification from the P2 Laplacian spectrum, and proposes ARIA as a candidate implementation-level testbed with hypothesised photon and microtubule specialisations. It does *not* deliver dynamical energy, linear-momentum, or gauge currents; those remain open items (§10).

1.2 Structure

§2 states $\mathcal{F}[\psi]$ explicitly and enumerates its two exact symmetries, with non-symmetries explicitly excluded. §3 records the targeted discrete continuity law on V_{600} as a programme hypothesis (not proved here), with Paper XVIII’s continuum continuity as the imported motivation. §4 explains why Noether’s theorem does not apply to the dissipative closure dynamics in the strict

variational sense, records the one conditional conservation statement (Proposition 4.1) and the H_4 -equivariance remark (Remark 4.5), and lists currents that are explicitly *not* claimed (energy, linear momentum, continuous angular momentum). §5 presents a heuristic Schrödinger-limit dispersion identification (Hypothesis 5.1) derived from the P2 §6 Laplacian spectrum, flagged as heuristic, not a theorem. §6 proposes a dictionary (Remark 6.1) between transport-law objects and the ARIA cognitive substrate, pending repository-backed verification. §7 states a photon-identification hypothesis (Hypothesis 7.1) conditional on the T_{PH} chain of [14]. §8 reframes microtubule transport as an ARIA-class hypothesis, explicitly *not* a first-principles derivation of the 13-fold symmetry. §9 cross-checks against Papers XII–XXXIV under their own stated scopes. §10 lists the five open items. §11 reports the delivered numerical witnesses (`verify_u1_conservation.py` and `verify_photon_dispersion.py`, machine-precision pass). §12 summarises.

1.3 Notation and conventions

V_{600} is the 600-cell vertex set (120 unit icosians, P2 §6). $\mathbf{A}_k$ is the Euclidean shell- k adjacency matrix. $L = 12I - \mathbf{A}_1$ is the combinatorial Laplacian. $\rho : V_{600} \times \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ denotes a density (conservation is *conditional* on Hypothesis 3.1 plus current antisymmetry; see Proposition 4.1). $j_{v \rightarrow w} : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}$ is the candidate flux along the oriented edge (v, w) . $\mathcal{F} : \mathbb{C}^{V_{600}} \rightarrow \mathbb{R}$ is the closure functional of Papers XII, XVIII, and XXII (realised on the $V_{600} \rightarrow \mathbb{C}$ finite-graph fields used throughout this paper). W is the learned coupling-weight tensor of ARIA.

2 The closure functional and its symmetries

Definition 2.1 (Closure functional). Following the setup of Papers XII and XXII, the closure functional on a square-integrable complex-valued field $\psi : V_{600} \rightarrow \mathbb{C}$ is

$$\mathcal{F}[\psi] = \alpha R[\psi] + \beta E[\psi] - \gamma C[\psi], \quad (1)$$

where

$$R[\psi] = \frac{1}{2} \sum_{v \in V_{600}} |\psi_v|^2 \log |\psi_v|^2, \quad (\text{residual / entropy term; } 0 \log 0 := 0) \quad (2)$$

$$E[\psi] = \frac{1}{2} \sum_{v \sim w} |\psi_v - \psi_w|^2, \quad (\text{edge-disagreement / kinetic term}) \quad (3)$$

$$C[\psi] = \frac{1}{2} (\psi, L_C \psi)_{\text{Herm}}, \quad (\text{quadratic closure term; Hermitian inner product}) \quad (4)$$

with coefficients $\alpha, \beta, \gamma \in \mathbb{R}$. We require L_C to be a real self-adjoint operator on $\mathbb{C}^{V_{600}}$ lying in the real polynomial algebra generated by the shell-adjacency matrices $\{\mathbf{A}_1, \dots, \mathbf{A}_8\}$ of P2 §6 [12]; this places L_C in the commutant of the H_4 -action and ensures $\mathcal{F}[\psi] \in \mathbb{R}$. Below we reserve the symbol Q for the total probability $\sum_v |\psi_v|^2$ (Proposition 4.1); the third functional term is denoted $C[\psi]$ throughout to avoid collision.

Remark 2.2. Definition 2.1 adopts an entropy–kinetic–quadratic decomposition inspired by the closure-functional language used across Papers XII, XVIII, and XXII [4, 6, 8]. The specific three-term form $\alpha R + \beta E - \gamma C$ is our working notation for the purposes of this paper; it is not a direct quotation of any single Proposition in those papers. The coefficients α, β, γ are left unspecified here; no local source in this paper fixes them.

Status of $\mathcal{F}$ in the present paper. In this paper, $\mathcal{F}$ functions as a *modelling functional* organising the transport hypotheses (the symmetry catalogue of Proposition 2.3, the continuity law of Hypothesis 3.1, and the dispersion identification of Hypothesis 5.1). It is not derived here from a more fundamental action principle, and the coefficient calibration is not supplied. Theorems 4.3 and 5.3 below do not depend on the $\mathcal{F}$ coefficients: they are stated directly for the Schrödinger-limit operator $i\dot{\psi} = H\psi$ with H in the shell-adjacency commutant.

Proposition 2.3 (Two exact symmetries of $\mathcal{F}$). *The closure functional (1) is invariant under the following two symmetry actions:*

(S1) *The continuous global $U(1)$ phase action: $\psi \mapsto e^{i\theta}\psi$, $\theta \in [0, 2\pi)$.*

(S2) *The finite H_4 Coxeter action: for each g in the (finite) Coxeter group H_4 , $\psi_v \mapsto \psi_{g^{-1}v}$.*

(S1) is a one-parameter Lie-group action; (S2) is a finite-group action with no Lie algebra — in particular, no continuous rotation generators — and therefore does not yield an infinitesimal Noether current in the classical sense. Any H_4 consequence used below is conditional, and amounts at most to the preservation of isotypic subspaces of $\mathbb{C}^{V_{600}}$ under commuting operators (and, under unitary evolution such as the Schrödinger-limit reduction of Paper XVIII, preservation of the norms of their projections); see Remark 4.5.

Proof. Use the standard Hermitian inner product $(\phi, \psi) = \sum_{v \in V_{600}} \overline{\phi_v} \psi_v$ on $\mathbb{C}^{V_{600}}$.

(S1). $R[\psi]$ depends on ψ only through $|\psi|$, hence is phase-invariant. $E[\psi]$ is invariant because $|e^{i\theta}\psi_v - e^{i\theta}\psi_w| = |\psi_v - \psi_w|$. $C[\psi] = \frac{1}{2}(\psi, L_C\psi)$ is invariant because $(e^{i\theta}\psi, L_C e^{i\theta}\psi) = e^{-i\theta} e^{i\theta} (\psi, L_C\psi) = (\psi, L_C\psi)$, using that L_C is $\mathbb{C}$ -linear and the Hermitian pairing absorbs the global phase.

(S2). $R[\psi]$ and $E[\psi]$ are sums over vertex-labels and edge-labels permuted by the H_4 -action, hence invariant. For $C[\psi]$, write U_g for the unitary induced on $\mathbb{C}^{V_{600}}$ by $v \mapsto g^{-1}v$ (P2 §6 [12]): since L_C lies in the real polynomial algebra generated by the shell-adjacency matrices (Definition 2.1), and each $\mathbf{A}_k$ commutes with U_g by the class-sum-centrality corollary of P2 §6, $U_g^\dagger L_C U_g = L_C$. Therefore $C[U_g\psi] = \frac{1}{2}(U_g\psi, L_C U_g\psi) = \frac{1}{2}(\psi, U_g^\dagger L_C U_g\psi) = C[\psi]$. $\square$

Remark 2.4 (Non-symmetries explicitly excluded). For completeness we record transformations that might appear to be symmetries but are not used here:

- *Time translation.* The closure gradient flow $\dot{\psi} = -\nabla\mathcal{F}$ is autonomous, but gradient flow dissipates $\mathcal{F}$. Energy-in-the-functional-sense is therefore not conserved under the closure gradient flow. In the Schrödinger limit of the Madelung pair of Paper XVIII [6], the usual QM statement applies *provided $\mathcal{H}$ is time-independent and self-adjoint* (as assumed in Hypothesis 5.1): the expectation $\langle \mathcal{H} \rangle_\psi$ is then conserved under the $i\dot{\psi} = \mathcal{H}\psi$ evolution. That expectation is a function of $\mathcal{H}$, not of $\mathcal{F}$ itself.
- *Binary-icosahedral left-regular action.* The $2I$ left-regular action is (S2) restated in group-algebra language; it is not independent.
- *Closure-scaling $\psi \mapsto \lambda\psi$.* Changes E by $|\lambda|^2$, Q by $|\lambda|^2$, and R by a mixed multiplicative/additive term; not a symmetry of $\mathcal{F}$. Restricted to a fixed L^2 -sphere it becomes a gauge, which we do not use.
- *σ -Galois involution and $SO(4)$ rotation.* Approximate symmetries of V_{600} (no quantitative residual bound is claimed here). If these enter the discrete continuity of §3 as source terms, they would contribute alongside the nonlinear closure residual of Paper XIX [7]; a precise identification with a specific bound from Paper XIX is not established here.

Only (S1) and (S2) are used downstream. This paper therefore enumerates two exact symmetries, not the larger count that would be needed for a four-current Noether catalogue; see the current discussion in §4.

3 The master continuity equation

3.1 Discrete continuity on V_{600}

Let $\rho_v(t) := |\psi_v(t)|^2$ be the density at vertex $v \in V_{600}$.

Hypothesis 3.1 (Programme target: discrete continuity analogue of Paper XVIII). Paper XVIII [6] establishes the exact continuum continuity equation $\partial_t \rho + \nabla \cdot (\rho v) = 0$ in configuration space under the Nelson-paired closure dynamics, with current velocity $v = -\nabla \mathcal{F} - (\sigma^2/2)\nabla \log \rho$. The discrete-substrate analogue — a vertex-wise continuity

$$\partial_t \rho_v + \sum_{w \sim v} j_{v \rightarrow w} = 0 \quad (v \in V_{600}) \quad (5)$$

for $\rho_v(t) := |\psi_v(t)|^2$, with an edge current $j_{v \rightarrow w}$ yet to be specified — is stated as a *programme target* of the present paper. The paper does not establish it; it records the target as the discrete shape that the Paper XVIII continuum continuity should take on V_{600} , and uses it downstream under that hypothesis. A precise discrete-master-equation derivation, correctly tracking the distinction between configuration-space and vertex-wise densities, is deferred to future work and is listed as open item (5) of §10.

3.2 Continuum limit (conjectural)

Hypothesis 3.2 (Conjectural long-wavelength continuum limit). Let $\rho(x, t)$ and $j(x, t)$ denote continuum fields obtained from $\rho_v, j_{v \rightarrow w}$ by a graph-to-continuum coarse-graining of P7-type (cascade-hydrodynamic-projection [13]). We conjecture that in the long-wavelength limit,

$$\partial_t \rho + \nabla \cdot j = \sigma_{\text{source}}, \quad (6)$$

with σ_{source} a residual source from the approximate symmetries of Remark 2.4. The precise Γ -convergence statement, including the identification of the source term, is open item (2) of §10; the present paper does not establish it. We use the symbol σ_{source} for the source term to avoid collision with the diffusion coefficient σ used elsewhere in the closure Langevin picture.

4 Symmetry-associated conservation statements (Noether caveats)

4.1 Noether caveat

Noether’s theorem in its strict form requires a variational action principle: the dynamics must be derivable from $\delta S = 0$ for some action functional S whose invariance under a one-parameter group yields a conserved current. The closure dynamics considered here is *dissipative* gradient flow plus Nelson pairing [4, 6], not variational in that strict sense; Paper XVIII’s Madelung reformulation rewrites the resulting dynamics as a coupled continuity / modified-Hamilton–Jacobi pair, but the pair arises from the Fokker–Planck operator, not from a first-principles action.

We therefore do *not* derive Noether currents in the classical variational sense. We record two symmetry-associated current candidates aligned with the two exact symmetries of Proposition 2.3. The $U(1)$ candidate is conserved *only* under Hypothesis 3.1 plus current antisymmetry (Proposition 4.1); the H_4 statement reduces in the Schrödinger limit to conditional block-norm invariance (Remark 4.5). Neither is a Noether current derived from a closed variational principle; that derivation is deferred.

Proposition 4.1 (Total-probability conservation under Hypothesis 3.1). *Suppose Hypothesis 3.1 holds and the discrete edge current $j_{v \rightarrow w}$ satisfies the natural antisymmetry $j_{w \rightarrow v} = -j_{v \rightarrow w}$. Then the total probability $Q(t) := \sum_{v \in V_{600}} |\psi_v(t)|^2$ is conserved along any evolution satisfying Hypothesis 3.1.*

Proof. Sum Equation (5) over v ; each edge (v, w) contributes $j_{v \rightarrow w} + j_{w \rightarrow v} = 0$ by the stated antisymmetry. $\square$

Remark 4.2. The antisymmetry hypothesis $j_{w \rightarrow v} = -j_{v \rightarrow w}$ is natural if the transport is driven by a difference of scalar quantities (as in Fick’s law or Fokker–Planck), and fails if the current has a non-antisymmetric drift component. Establishing the antisymmetry directly from the closure Langevin generator is part of the programme target of Hypothesis 3.1.

Theorem 4.3 (Discrete continuity + $U(1)$ conservation in the Schrödinger limit, with explicit current). *On the Schrödinger limit $i\dot{\psi} = H\psi$ of the Madelung pair (Paper XVIII [6]), where H is the V_{600} graph Laplacian (or any real-symmetric nearest-neighbour-supported operator on $\mathbb{R}^{V_{600}}$, i.e. $H_{vw} \neq 0 \Rightarrow v \sim w$ or $v = w$), define the explicit edge current*

$$j_{v \rightarrow w} := -2 \operatorname{Im}(\overline{\psi_v} H_{vw} \psi_w) \quad \text{for each oriented adjacent pair } (v, w). \quad (7)$$

Then:

1. The current is antisymmetric: $j_{w \rightarrow v} = -j_{v \rightarrow w}$ (algebraic consequence of $H_{wv} = H_{vw} \in \mathbb{R}$).
2. The pointwise discrete continuity equation $\dot{\rho}_v + \sum_{w \sim v} j_{v \rightarrow w} = 0$ holds at every $v \in V_{600}$ and every t , as an algebraic identity following from $\dot{\rho}_v = +2 \operatorname{Im}(\overline{\psi_v} (H\psi)_v)$.
3. Hence Hypothesis 3.1 is realised exactly (not approximately) by the explicit current (7), and Proposition 4.1 promotes from conditional to unconditional in the Schrödinger limit on $H = L$: $Q(t) = \sum_v |\psi_v(t)|^2$ is conserved exactly.

Restriction to nearest-neighbour-supported H is essential: the continuity sum is over $w \sim v$ only, so a H with non-nearest-neighbour matrix entries (e.g. a polynomial $\mathbf{A}_2 = \text{shell-2 adjacency}$, which is in the shell-adjacency commutant but not nearest-neighbour) requires extending the current to all pairs with $H_{vw} \neq 0$ and the corresponding extended-graph continuity sum. The graph Laplacian $L = D - A$ on V_{600} is nearest-neighbour-supported by construction.

Proof. For H real symmetric, $j_{v \rightarrow w} = -2 \operatorname{Im}(\overline{\psi_v} H_{vw} \psi_w)$ and $j_{w \rightarrow v} = -2 \operatorname{Im}(\overline{\psi_w} H_{wv} \psi_v) = -2 \operatorname{Im}(\overline{(\overline{\psi_v} H_{vw} \psi_w)}) = +2 \operatorname{Im}(\overline{\psi_v} H_{vw} \psi_w) = -j_{v \rightarrow w}$, giving (1). For (2): $i\dot{\psi} = H\psi \Rightarrow \dot{\psi} = -iH\psi$, so $\dot{\rho}_v = \partial_t |\psi_v|^2 = 2 \operatorname{Re}(\overline{\psi_v} \dot{\psi}_v) = 2 \operatorname{Re}(-i \overline{\psi_v} (H\psi)_v) = +2 \operatorname{Im}(\overline{\psi_v} (H\psi)_v)$. The expansion of $(H\psi)_v = \sum_w H_{vw} \psi_w = H_{vv} \psi_v + \sum_{w: w \neq v, H_{vw} \neq 0} H_{vw} \psi_w$ contributes a diagonal piece $\overline{\psi_v} H_{vv} \psi_v = H_{vv} |\psi_v|^2 \in \mathbb{R}$ which has zero imaginary part and so drops out; the remaining off-diagonal sum (restricted to nearest neighbours $w \sim v$ by the nearest-neighbour-support hypothesis on H) gives $\dot{\rho}_v = \sum_{w \sim v} 2 \operatorname{Im}(\overline{\psi_v} H_{vw} \psi_w) = -\sum_{w \sim v} j_{v \rightarrow w}$. (3) follows from (1)+(2) and Proposition 4.1.

An independent numerical witness — evolution under $U = e^{-iLt}$ for 200 steps on a normalised random complex initial state, with the analytic continuity residual measured at machine precision — is performed by `repro/verify_u1_conservation.py`; see Table in §11. Result: Q -drift $\sim 3 \times 10^{-15}$, pointwise continuity residual $\sim 3 \times 10^{-16}$, current antisymmetry exactly 0. $\square$

Remark 4.4 (What Theorem 4.3 closes and what it does not). Theorem 4.3 closes the discrete continuity hypothesis (Hypothesis 3.1) and Proposition 4.1 in the Schrödinger limit of Paper XVIII — which is precisely the regime in which the dispersion identification of Hypothesis 5.1 also lives. It does *not* close the same statements for the full *dissipative* closure dynamics $\dot{\psi} = -\nabla \mathcal{F} + \text{noise}$: gradient-flow $\mathcal{F}$ -descent does not in general preserve $Q = \|\psi\|^2$ (see Remark 4.6), so the discrete continuity equation in that regime requires either a different current or a different conservation law (e.g. a $\mathcal{F}$ -monotone law). The Schrödinger-limit case is the closure-dynamics regime in which the conditional probability-current layer of this paper is honestly closed.

Remark 4.5 (H_4 -equivariance: what it actually gives, and what it does not). H_4 is a finite Coxeter group, not a Lie group, so it has no Lie algebra and no infinitesimal generators: the familiar “angular momentum current” from a continuous rotation group does not apply directly

to (S2). What (S2) gives, algebraically, is a unitary representation $\widehat{g}$ on $\mathbb{C}^{V_{600}}$ induced by the permutation $v \mapsto g^{-1}v$ (P2 §6 [12]), and the fact that the shell-adjacency operators $\mathbf{A}_k$ lie in the commutant of this representation. The commutant property has two concrete consequences:

1. Every polynomial in the shell-adjacency operators preserves the $2I$ -isotypic subspaces of $\mathbb{C}^{V_{600}}$. This is a statement about *subspaces*, not about individual projected vectors: a vector inside an isotypic subspace stays inside that subspace under any shell-adjacency-algebra time evolution, but its direction within the subspace is not constrained.
2. *Conditional on* (a) the Schrödinger-limit reduction $i\dot{\psi} = \mathcal{H}\psi$ of the Madelung pair of Paper XVIII [6], *and* (b) the additional hypothesis that $\mathcal{H}$ is itself built from the shell-adjacency algebra (so that it lies in the commutant of the H_4 -action), the unitary evolution $e^{-i\mathcal{H}t}$ preserves the norm of each isotypic projection $\|P_\rho\psi\|$. *Neither* condition is established here: (a) is the Schrödinger-limit hypothesis already carried by Hypothesis 5.1, and (b) requires the linearised variation of $\mathcal{F}$ around a closure equilibrium to respect the H_4 -symmetry, which in turn requires the equilibrium itself to respect H_4 . Without (b) the linearisation in general mixes isotypic components.

A continuous angular-momentum current would emerge only in the continuum $\text{SO}(4)$ -limit after the Γ -convergence of open item (2) of §10; we do not attempt that limit here. Under the full (dissipative) closure dynamics, even the conditional block-norm preservation of item 2 is at best approximate; a precise bound on the drift would require combining Paper XII's gradient-flow dissipation [4] with the nonlinear residual of [7], which is not attempted here.

Remark 4.6 (On "energy" and "linear momentum"). Two currents that might be expected here but are not present:

- *Energy.* Gradient flow dissipates $\mathcal{F}$; energy in the functional sense is therefore not conserved. The standard QM conservation statement under Schrödinger evolution — that $\langle \mathcal{H} \rangle$ is conserved under $i\dot{\psi} = \mathcal{H}\psi$ for time-independent self-adjoint $\mathcal{H}$ — applies to the linearised Hamiltonian $\mathcal{H}$, which is not $\mathcal{F}$.
- *Linear momentum.* V_{600} is finite and bounded; there is no $\mathbb{R}^4$ -translation symmetry to give linear momentum.
- *Continuous angular momentum.* H_4 is a finite Coxeter group; it has no continuous rotation generators. A continuous angular-momentum current would emerge only after the Γ -limit to the continuum $\text{SO}(4)$ -invariant theory (open item (2) of §10).

None of these is claimed in this paper.

Remark 4.7 (Charge current: conditional-only, and $\text{SU}(5)$ assignment disclaimer). Conditional on Hypothesis 3.1 (the discrete continuity law) and on the antisymmetry $j_{w \rightarrow v} = -j_{v \rightarrow w}$ of the associated current, the $U(1)$ -phase-rotation charge $Q = \sum_v |\psi_v|^2$ is conserved on V_{600} (Proposition 4.1). For the Schrödinger-limit case $H = L$ this is now *unconditional* via Theorem 4.3 (which delivers both the continuity equation and the antisymmetric current explicitly); the Hypothesis 3.1 / antisymmetry conditioning still applies to the full *Langevin-with-noise* closure dynamics, where the question is part of open programme item (5) of §10. We do *not* claim in this paper that Paper XXII [8] contains a theorem-level derivation of the fractional $\text{SU}(5)$ charge assignments; local verification against `paper-xxii.tex` does not surface such a derivation, and `docs/gaps.md` records the fractional quark charges as an open gap. A first-principles identification of SM representation labels with $2I$ -isotypic sectors of P2 §6 is not attempted here.

5 Dispersion on the 600-cell

Hypothesis 5.1 (Heuristic Schrödinger-limit dispersion identification). Assume $\beta > 0$ (so that the edge-disagreement term in $\mathcal{F}$ is a positive kinetic cost) and that, in the Schrödinger limit of the Madelung pair of Paper XVIII [6], the closure dynamics reduces to $i\dot{\psi} = \mathcal{H}\psi$ for some *time-independent self-adjoint* operator $\mathcal{H}$ built from the shell-adjacency algebra of P2 §6 [12]. Both conditions are additional hypotheses here: $\mathcal{H}$ time-independence is needed for the conservation-of-expectation language used in §4.1; the shell-adjacency form of $\mathcal{H}$ is *not* imported from Paper XVIII. Under these hypotheses we *adopt the formal ansatz* that the long-wavelength dispersion of each isotypic block ρ is

$$\omega^2 = c_{\text{eff}}^2 k^2 + m_\rho^2, \quad (8)$$

where $m_\rho^2 := \beta \lambda_\rho$, λ_ρ is the Laplacian eigenvalue on block ρ , $c_{\text{eff}}^2 = \beta a^2$, and a is the 600-cell edge length. The P2 spectrum

$$\lambda_\rho \in \{0, 9 - 3\sqrt{5}, 10 - 2\sqrt{5}, 9, 12, 14, 10 + 2\sqrt{5}, 15, 9 + 3\sqrt{5}\}$$

is used as the eigenvalue data [12]. **This is a heuristic physical identification, not a theorem:** (i) the reduction $\dot{\psi} = -\nabla \mathcal{F} + \text{noise} \Rightarrow i\dot{\psi} = \mathcal{H}\psi$ is not derived here (inherited from Paper XVIII's Madelung pair conditional on its hypotheses); (ii) the notion of a long-wavelength wavenumber k on the finite graph V_{600} is itself a coarse-graining construct tied to the continuum Γ -limit (open item (2) of §10); (iii) the unit convention $m_\rho^2 = \beta \lambda_\rho$ with $\beta a^2 = c^2$ is an external calibration chosen to match the relativistic-dispersion form, not derived. Each of these three items is a hypothesis in its own right.

Remark 5.2 (Spectral sectors under the heuristic identification). If Hypothesis 5.1 is adopted, the nine Laplacian eigenvalues of P2 §6 split into five $\mathbb{Q}$ -rational values $\{0, 9, 12, 14, 15\}$ and four irrational values $\{9 \pm 3\sqrt{5}, 10 \pm 2\sqrt{5}\}$. The $\lambda = 0$ eigenmode (trivial isotypic component) corresponds to $m^2 = 0$ and, if it is physical, to the massless photon sector discussed in §7 under the additional photon-identification hypothesis. The four nontrivial rational sectors give four rational-irrep mass-squared values which reuse the same four rational eigenvalues singled out arithmetically in Paper V [3]; note that Paper V uses $\{9, 12, 14, 15\}$ as ingredients in a particle-mass numerical correspondence and does *not* itself establish a dispersion-based mass-squared sector identification. The four irrational sectors are additional spectral content not used by Paper V's 13-particle table.

Theorem 5.3 (Photon-sector spectral witness on V_{600}). *On the 600-cell graph Laplacian L with spectrum (multiplicities) $\{0 : 1, 9 - 3\sqrt{5} : 4, 10 - 2\sqrt{5} : 9, 9 : 16, 12 : 25, 14 : 36, 10 + 2\sqrt{5} : 9, 15 : 16, 9 + 3\sqrt{5} : 4\}$:*

1. *The $\lambda = 0$ eigenspace is one-dimensional, spanned by the uniform vector $u_0 \propto (1, \dots, 1)$.*
2. *The trivial-sector projector $P_0 = u_0 u_0^\top$ satisfies $L P_0 = 0$ exactly (algebraic; numerical witness $\|L P_0\|_F \lesssim 10^{-13}$).*
3. *Under the Schrödinger-limit evolution $i\dot{\psi} = L\psi$, an initial state $\psi_0 \in \text{range}(P_0)$ is exactly stationary: $\psi(t) = \psi_0$ for all t (zero-frequency mode).*
4. *For any non-trivial eigenmode u_λ with eigenvalue $\lambda > 0$, the evolution is $\psi(t) = e^{-i\lambda t} u_\lambda$, i.e. the per-mode angular phase rotates at rate λ (equivalently, the per-mode angular frequency in the Schrödinger-limit sense is $\omega = \lambda$).*

The connection to the dispersion ansatz $\omega^2 = c_{\text{eff}}^2 k^2 + m_\rho^2$ of Hypothesis 5.1 is not delivered here: *that ansatz invokes a long-wavelength wavenumber k (a coarse-graining construct tied to*

the open Γ -limit, item (2) of §10) and a unit-conversion calibration $m_\rho^2 = \beta\lambda_\rho$ (the heuristic identification of Hypothesis 5.1, also conditional). Theorem 5.3 verifies the substrate-level facts about the trivial / non-trivial spectral sectors of L ; the gauge-side photon identification (the T_{PH2} chain of [14], marked “needs polish”) and the dispersion identification remain hypotheses. The trivial-sector stationarity is the substrate-side content of the massless ($m = 0$) physical interpretation, not a derivation of it.

Proof. (1)+(2) follow from the standard fact that the Laplacian of a connected regular graph has a one-dimensional kernel spanned by the constant vector. (3)+(4) follow from the spectral theorem applied to L . The Schrödinger evolution on each eigenspace is $u_\lambda(t) = e^{-i\lambda t}u_\lambda(0)$; for $\lambda = 0$ this reduces to $u_0(t) = u_0(0)$.

A complete numerical verification — spectrum check, projector annihilation $\|LP_0\|_F < 10^{-13}$, trivial-sector stationarity to machine precision, non-trivial frequency match to $\sim 10^{-15}$ — is performed by `repro/verify_photon_dispersion.py`; see Table in §11. $\square$

Remark 5.4 (Theorem 5.3 as the substrate-side photon witness). Theorem 5.3 verifies the substrate-level content of the photon-identification hypothesis (Hypothesis 7.1): the $\lambda = 0$ massless mode exists, is one-dimensional, lies in the trivial $2\mathbf{I}$ -isotypic sector, and is preserved exactly under the Schrödinger-limit evolution. It does *not* close Hypothesis 7.1: the T_{PH2} descent claim of [14] (the gauge-side identification of the trivial mode with the diagonal $U(1)_{\text{EM}}$) remains marked “needs polish” in the source note, and the local-gauge-current reinterpretation is not delivered here.

Remark 5.5 (Unit conventions). The transport law is stated with $\hbar = 1$. Laplacian eigenvalues λ_ρ are dimensionless integers (or $\mathbb{Q}(\sqrt{5})$ -valued for irrational sectors); mass-squared values $m_\rho^2 = \beta\lambda_\rho$ carry the dimensions of the β -coupling. In physical units, $\beta a^2 = c^2$ fixes the speed of light, and m_ρ^2 becomes mass-squared in units of β . Absolute mass values require the Paper V calibration convention (shell index $|S| = 5$; mass ratios fixed by $R_D(4) = 15$ of P2 §6); that convention is an input to the transport law, not derived from it.

6 ARIA as a proposed verification-target testbed (not done here)

This section proposes the ARIA cognitive system (`aria-chess` repository, paper [2]) as a *candidate* implementation-level testbed for the transport-law framework of §§3–5. The published `aria-chess` preprint reports a *geometry-fixed substrate witness* on V_{600} under the closure-response operator $C_\varphi = L_M + \varphi^{-2}I$ [1]; learned coupling weights and an $E_8 \oplus V_{600}$ product structure are *not* the active substrate of that paper, but rather features of the release-bundle kernel scripts (`kernel/lyapunov_selector.py`, `kernel/self_model_stream.py`). Treating those kernel artifacts as a *proposed* implementation-level testbed for the present transport-law framework is the load that this section actually carries. The dictionary in Remark 6.1 hypothesises an object-level correspondence, but the correspondence is not proved here and no repository traces are analysed against the transport observables (Fick / Nernst-Planck / cascade-pressure balance) in the present draft. Transport-observable verification is listed as open item (4) of §10.

6.1 Mapping

Remark 6.1 (Proposed ARIA–transport-law dictionary). We propose — without proof, and without repository-backed verification at the time of writing — the following object-level correspondence between the transport-law framework of §§2–5 and the ARIA implementation:

Transport-law object	ARIA realisation (proposed)
Density ρ_v (conservation only under Hypothesis 3.1)	cascade pressure at slot v
Flux $j_{v \rightarrow w}$	<code>couple_tick</code> exchange on edge (v, w)
Conductivity tensor	learned coupling weight W_{vw}
Dirichlet boundary	homeostatic reset
Source / sink	external input / decay-reset
Thermodynamic class	non-equilibrium steady state (NESS)
Passive regime	linear <code>couple_tick</code> (Fick-type)
Active regime	nonlinear drift along ∇W (Nernst-Planck-type)

Nothing in the table is established in this paper. Each row is a proposed correspondence pending (i) a generator-level derivation of the discrete continuity (open item (5)) and (ii) repository-level verification against actual `aria-chess couple_tick` traces (open item (4)).

Remark 6.2 (Proposed passive / active regimes). The passive / active split of the proposed ARIA dictionary is hypothesised to align with the VFD linear / nonlinear closure-dynamics distinction:

- **Passive (linear, hypothesised)** = near-equilibrium regime of Paper XVIII’s Madelung pair; W frozen; Fick-type linear transport.
- **Active (nonlinear, hypothesised)** = regime where Paper XIX’s closure residual becomes non-negligible; W co-evolves with ρ ; selection appears; Nernst-Planck-type transport.

The identification of ”selection” with ”closure residual non-negligible” is hypothesised, not proved. The passive / active phase diagram is presented here as a target for empirical characterisation, not as a derived result.

6.2 Objectives 2, 4, 6 as proposed ARIA verification targets

Under the proposed ARIA dictionary of Remark 6.1, three of the objectives stated in the WO for this paper become empirically-testable *targets* — items to be verified against actual `aria-chess couple_tick` trace data. None of the three is established in this paper; each would require the continuity hypothesis 3.1 to be closed *and* the repository-level traces to be analysed against the transport-law equations. We state the targets explicitly:

- **Objective 2 (Noether-type charge) target.** Under the dictionary, ARIA’s total cascade pressure (if conserved up to explicit source/sink accounting) *would* be identified with the $U(1)$ -phase charge $Q = \sum_v |\psi_v|^2$ of Proposition 4.1, and the `couple_tick` exchange *would* be identified with the current $j_{v \rightarrow w}$ of Hypothesis 3.1. The target for empirical verification is: does the total ARIA pressure remain conserved (up to explicit source/sink accounting) along realistic `couple_tick` trajectories?
- **Objective 4 ($T^{\mu\nu}$) target.** Under the dictionary, the ARIA NESS *would* be hypothesised to satisfy a zeroth-component balance $\sum_v (\text{pressure flux})_v = \text{input} - \text{decay}$, analogous to (but distinct from) the centering identity of Paper XXVII [9]. The target is to test this balance empirically; no identification with a relativistic $T^{\mu\nu}$ is claimed.
- **Objective 6 (Fick / Nernst-Planck) target.** The target is to test whether `couple_tick` in the passive (frozen- W) regime exhibits Fick-type behaviour $j \approx -D\nabla\rho$ and whether the active (learning) regime exhibits a Nernst-Planck-type correction $j \approx -D\nabla\rho + \mu\rho\nabla W$, with D, μ extracted from ARIA rather than fitted here.

None of the three targets is demonstrated in the present paper; all three are listed as part of open item (4) of §10.

6.3 Numerical verification plan

The numerical verification of §11 delivers two scripts that establish substrate-level facts (Schrödinger-limit $U(1)$ conservation; photon-sector spectral witness on V_{600}). It does *not* run `aria-chess couple_tick` traces against any of the three Fick / Nernst-Planck / cascade-pressure-balance targets above; analysing the published `aria-chess` preprint [2] against those specific transport observables remains part of open item (4) of §10.

7 Massless zero mode as a conditional photon-sector witness

This section records a speculative conditional note on how a photon sector might sit inside the transport-law framework. The cumulative hypothesis load is heavy and includes the $T_{\text{PH}2}$ descent claim of [14], which is not closed in the repository. Readers should treat what follows as a conditional identification proposal, not a near-theorem:

Hypothesis 7.1 (Conditional photon-identification target). Under the cumulative hypothesis load {3.1, 5.1, the T_{PH} chain of the internal note [14]} — in particular $T_{\text{PH}2}$, the descent claim for the phason translation group T of L_{12} (written as $\mathbb{Z}[\varphi]^4$ in the source note; the later LaTeX papers [11, 15] adopt the $\mathbb{Z}[\varphi]^2$ convention together with an asserted rank-convention reconciliation, but no translated proof of $T_{\text{PH}2}$ is carried out in the repository) onto the diagonal $U(1)_{\text{EM}}$ at the H_4 rung, which the source note marks as “needs polish” at one place and “closed” at another, and which we cite only in the former sense — we *record the candidate identification* of the $\lambda = 0$ eigenmode of L on V_{600} with the photon sector. Under that identification, the massless $\lambda = 0$ mode lies in the trivial isotypic sector of the $2I$ decomposition, and its dispersion is $\omega = c_{\text{eff}}|k|$ (from Hypothesis 5.1 with $m = 0$). Any relation between the conditional global scalar charge $Q = \sum_v |\psi_v|^2$ of Proposition 4.1 and the local $U(1)_{\text{EM}}$ gauge current is a separate $T_{\text{PH}2}$ -dependent reinterpretation; the present paper does *not* derive a local electromagnetic current, and does *not* assign an electric charge to the photon here.

The present paper does *not* establish any of the following: (i) that this identification is unique; (ii) that a global scalar charge on V_{600} can be reinterpreted as a local $U(1)_{\text{EM}}$ gauge current derivable from the closure dynamics (open item (1) of §10); (iii) that the photon in this framework carries a local gauge potential A_μ . In ARIA terms: photons would correspond to the homogeneous-substrate limit $W = 0$.

8 Microtubule transport: reframed

The original VFD programme posited that microtubules realise the cascade substrate via the $T_{\text{MT}1}$ identification (`cascade-alpha-chain-complete-theorem.md`): “the biological chamber implementing the cascade’s 13D translation-selector function has cyclic symmetry C_{13} ”. That identification explicitly does not close rigorously; the source document records it as *structural conjecture*.

8.1 Reframed claim

Taking ARIA as a candidate active-transport substrate of the closure-dynamics class (Remark 6.1; a proposal pending repository-backed verification), the claim to be attached to microtubules is no longer a first-principles derivation of the number 13. It is instead:

Hypothesis 8.1 (Microtubule as ARIA-class active substrate). We hypothesise that biological microtubules may be active-transport substrates of the nonlinear regime of Remark 6.1: learned-conductivity anisotropy, NESS thermodynamics, nonlinear drift along ∇W . Under this hypothesis, the 13-fold helical symmetry would be a selected operating point of the active transport —

stabilised by whatever biological cost function selects it — rather than a cascade-combinatorial number derivable from L_{12} .

Remark 8.2. Hypothesis 8.1 is a tractable biophysics claim rather than a combinatorial-uniqueness theorem. It predicts: (i) the microtubule substrate should exhibit Fick’s law in the passive limit and Nernst-Planck in the active limit; (ii) anaesthetic disruption ($T_{\text{MT}5}$ in the source document) should operate by disrupting the learned- W anisotropy, not by perturbing the cascade directly; (iii) the helical pitch ($T_{\text{MT}4}$) is fixed by the biological cost function, not by the cascade geometry. None of these is proved here; all are open for empirical test.

9 Consistency checks with Papers XII–XXXIV

- **Paper XII (gradient flow)** [4]. Paper XII introduces both the unprojected gradient flow $\dot{\psi} = -\nabla\mathcal{F}$ and, on the admissible manifold, a projected version $\dot{\psi} = -\Pi(\nabla\mathcal{F})$ (paper-xii.tex, lines 87–99). Together with the Nelson pairing of Paper XVIII, that dynamics motivates the discrete continuity *hypothesis* (Hypothesis 3.1) stated here; the discrete equation is *not* derived in the present paper.
- **Paper XVI (WKB transport)** [5]. Paper XVI establishes WKB transport in a semi-classical regime. We note that WKB transport is a different formal framework from the Nelson-paired continuity targeted by Hypothesis 3.1, and do not claim a precise reduction in this paper.
- **Paper XVIII (Nelson pairing)** [6]. Paper XVIII’s continuum continuity Proposition (Proposition 3.1 at lines 115–130 of paper-xviii.tex; $\partial_t\rho + \nabla \cdot (\rho v) = 0$) is the continuum analogue of our Hypothesis 3.1. Paper XVIII does *not* prove the discrete version; the present paper records the discrete version as a programme target only. The Schrödinger-limit reduction we invoke in Hypothesis 5.1 is Paper XVIII’s exact-at-equilibrium, leading-order-near-equilibrium result, not a blanket reduction of the full closure dynamics.
- **Paper XIX (closure residual)** [7]. Paper XIX establishes a nonlinear residual in the Schrödinger-normalised closure dynamics and emphasises exact L^2 conservation; it does *not* establish a source term in a continuity equation. We therefore do *not* claim that Paper XIX’s residual provides a dissipative / source term of the transport law here. The informal analogy with “Nernst-Planck-type active transport” in §6 is a proposed interpretation specific to the ARIA dictionary and is not an attribution to Paper XIX.
- **Paper XXVII (discrete graph-Poisson)** [9]. Paper XXVII explicitly emphasises that its $\sum_v (S_v - \bar{S}) = 0$ is a compatibility / centering identity, *not* a dynamical conservation law. This paper does *not* claim that our Hypothesis 3.1 subsumes the XXVII identity; the two live in different formal regimes (field equation on event poset vs. transport equation on V_{600}).
- **Paper XXX (stationary measure and Born rule)** [10]. Paper XXX contains a Route C conditional stationary-measure / existence argument and a Route E conditional uniqueness argument; only the latter depends on the full completeness / basis-realisability / $U(N)$ -covariance / frame-function / continuity / eigenstate-certainty / $N \geq 3$ package. This paper does *not* claim a $t \rightarrow \infty$ fixed-point identification between our (5) and Paper XXX’s stationary measure under either route; such an identification would require invoking the relevant route’s hypotheses and is deferred.
- **Paper XXXIV (α -chain)** [11]. The sum $50 = 9 + 12 + 14 + 15$ that appears in Paper XXXIV is the sum of our m_ρ^2/β values on the four rational massive sectors (Remark 5.2). The 87 count of Paper XXXIV is an adopted convention under a Kostant-gauge

hypothesis, not a theorem-level slot count; we do not claim it as a physical slot count of the $U(1)$ current of Proposition 4.1.

10 Open items

Five open items separate the present paper from a closed theorem chain:

1. **Gauge-field emergence.** Gauging the global $U(1)$ of (S1) to a local $U(1)$ introduces an electromagnetic potential A_μ . Whether $\mathcal{F}[\psi]$ naturally carries a local phase gauge (so that A_μ is derived, not added) is open; the present paper works with the global charge current only.
2. **Γ -limit coarse-graining.** The continuum passage from V_{600} to a continuum hydrodynamic limit is sketched, not proved. A full proof requires a Γ -convergence argument with the small parameter being the ratio of the 600-cell edge length a to the coarse-graining scale; candidate machinery: [16]-style discrete-to-continuum passage.
3. **Learned W as gauge potential.** ARIA’s learned W plays the role of a classical, scalar-valued conductivity tensor. Whether W can be identified with a gauge potential in the quantised transport law is open; this is the ARIA-side analogue of Item 1.
4. **ARIA repository-level verification.** The ARIA dictionary of Remark 6.1 is a proposal pending verification against actual `aria-chess couple_tick` trace data. The numerical programme of §11 describes the required checks; they are not yet carried out.
5. **Discrete continuity derivation.** Hypothesis 3.1 records (5) as the discrete-substrate analogue of Paper XVIII’s continuum continuity. A precise generator-level derivation of the edge current $j_{v \rightarrow w}$ on V_{600} from the closure Langevin dynamics, including the antisymmetry property used in Proposition 4.1, is deferred.

11 Numerical verification

Two scripts in `papers/transport-law/repro/` verify the two new theorems of this paper. Both are self-contained (no external data; numpy + scipy only) and run in seconds on a laptop.

`verify_u1_conservation.py` — Theorem 4.3

Builds V_{600} + the graph Laplacian L , evolves a localised complex initial state ψ under $U = e^{-iL\Delta t}$ for 200 steps at $\Delta t = 0.05$, and at every step verifies (a) the analytic continuity residual $\dot{\rho}_v + \sum_w j_{v \rightarrow w}$ at every vertex, (b) total Q conservation, and (c) current antisymmetry $j_{v \rightarrow w} + j_{w \rightarrow v}$. Expected output (recorded in `results_u1_conservation.json`):

quantity	value
<code>n_steps</code>	200
<code>Q_initial, Q_final</code>	1.0, 1.0
<code>max_Q_drift</code> (over 200 steps)	$\sim 3 \times 10^{-15}$
<code>max_continuity_residual</code> (analytic)	$\sim 3 \times 10^{-16}$
<code>max_antisymmetry_violation</code>	exactly 0
<code>all_three_pass</code>	true

verify_photon_dispersion.py — Theorem 5.3

Builds L , computes the spectrum, verifies $\lambda_{\min} = 0$ with multiplicity 1 and uniform eigenvector, the operator-level zero $L P_0 = 0$, the trivial-sector stationarity, and the non-trivial-mode frequency $\omega = \lambda$. Expected output (recorded in `results_photon_dispersion.json`):

quantity	value
lam_min	$\sim 6 \times 10^{-15}$ (i.e. ≈ 0)
trivial_multiplicity	1
trivial_eigenvector_uniform_dev	$\sim 4 \times 10^{-16}$
L_P0_frobenius_norm	$\sim 1 \times 10^{-14}$
trivial_propagation_max_dev	$\sim 5 \times 10^{-15}$
non_trivial_freq_match_dev	$\sim 5 \times 10^{-15}$
all_six_pass	true

These two scripts are the substrate-level numerical witness for the two theorems delivered in this paper. The companion `aria-chess` preprint [2] ships a deterministic recurrent-layer trajectory (`kernel/lyapunov_selector.py`, `kernel/self_model_stream.py`) that realises the active regime of Remark 6.1 on the same V_{600} substrate at the level of an EEG-grade substrate-witness (6/6 drug/sleep signatures and 18/18 preregistered cortical correspondences against literature-derived thresholds; substrate-witness scope, not a derivation of consciousness). That artifact is *not* a verification of the Fick / Nernst-Planck / cascade-pressure-balance targets of §6: the trajectory does not extract D , μ , or test the source/sink balance of objectives 2/4/6 against the transport-law equations. Such transport-objective verification remains part of open item (4) of §10.

12 Conclusion

We have assembled a conditional transport framework for the closure dynamics on the 600-cell substrate of P2 §6. The framework contains: two exact symmetry identifications (Proposition 2.3: $U(1)$ phase and H_4 isometry); one conditional conservation statement (Proposition 4.1: total-probability conservation under Hypothesis 3.1 plus current antisymmetry); one Schrödinger-limit H_4 -equivariance remark recording a conditional isotypic-population-invariance statement (Remark 4.5, valid only under the Schrödinger-limit reduction *and* the additional hypothesis that the limit generator is itself H_4 -equivariant; *not* an angular-momentum current, because H_4 has no Lie algebra); and one heuristic dispersion identification tying the P2 spectrum to a relativistic dispersion form (Hypothesis 5.1). Photon and microtubule applications are stated as hypotheses conditional on the T_{PH} chain of [14] and on an ARIA-class reframing of microtubule transport, respectively.

The paper does not fit any numerical parameter; the coefficients α, β, γ of Definition 2.1 are left unspecified, with their calibration deferred to downstream papers that build on this framework. Five open items separate the present framework from a closed theorem chain: gauge-field emergence, Γ -limit coarse-graining, learned- W -as-gauge-potential, ARIA repository-level verification, and a generator-level discrete-continuity derivation. The paper is a hypothesis-audited framework with two Schrödinger-limit substrate theorems (Theorem 4.3 and Theorem 5.3), not a closed theorem chain across the full programme; it should be read accordingly.

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